Cosmological tests of gravity with galaxy clusters

Eleni Tsaprazi

Observatoire de Paris - PSL Fellow

Laboratory for the study of the Universe and eXtreme phenomena (LUX)

Journées Scientifiques Galaxies de l'ATCG 2026

Why gravity? Why clusters? Why now?

Fundamental questions remain unanswered

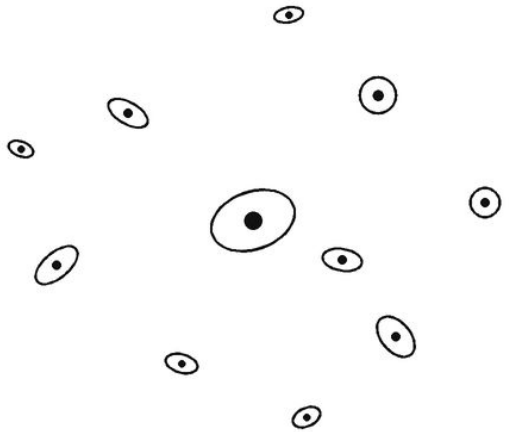
Galaxy clusters: complementary observables

Exploit potential: Stage-IV now, Stage-V ahead

Data quality: (i) control of systematic effects (ii) new probes

Multi-scale approach to modified gravity

Gravity at intracluster scales



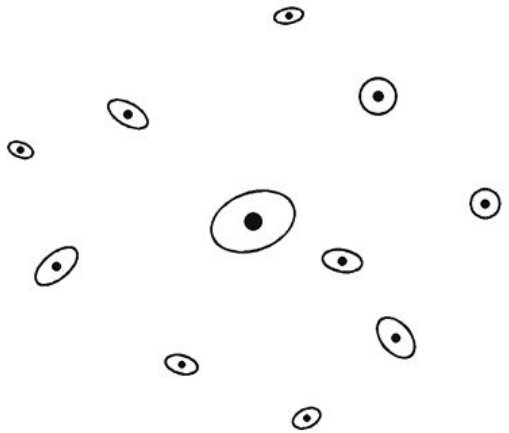
$$ds^2 = -(1+2\Psi)\,dt^2 + a^2(1-2\Phi)\,d\mathbf{x}^2$$

Gravitational redshifts

[kpc – Mpc]

$\Delta\Psi$

Gravity at cluster scales



$$ds^2 = -(1+2\Psi)\,dt^2 + a^2(1-2\Phi)\,d\mathbf{x}^2$$

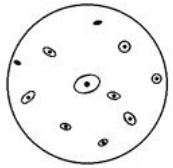
Gravitational wave strong lensing

[< 10 Mpc]

$\Phi + \Psi$

Gravity at intercluster scales

$$ds^2 = -(1+2\Psi)\,dt^2 + a^2(1-2\Phi)\,d\mathbf{x}^2$$



Clustering

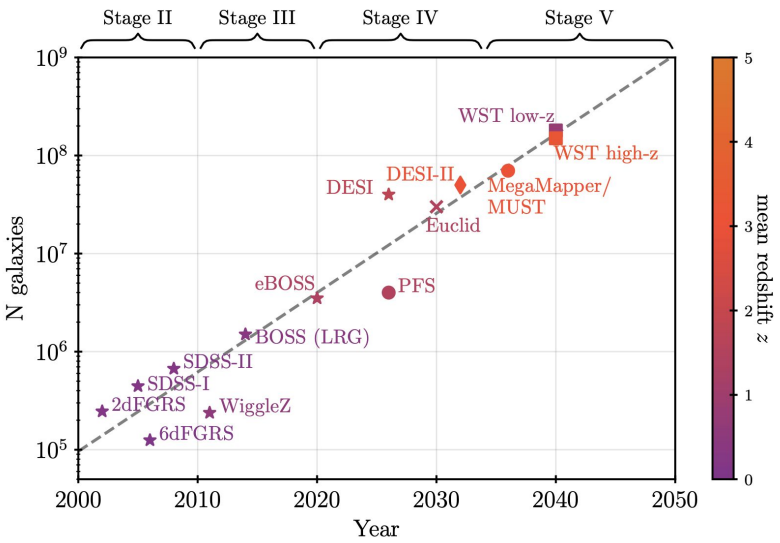
[> 10 Mpc]

$\nabla\ \Psi$

Gravitational redshifts: gravity at intracluster scales

Signal: φ_{grav} at galaxy - φ_{grav} at centre

- Direct test of gravity: φ(M)
- SDSS DR16 / DESI / Euclid
- **Widefield Spectroscopic Telescope**
  - Conceptual planning

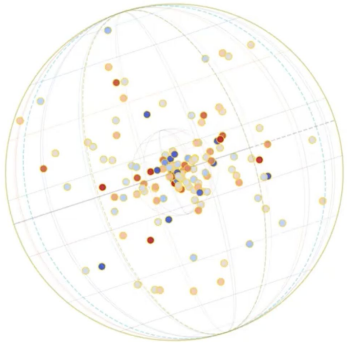


How do uncertainties impact the signal?

How to optimise survey design?

Tsaprazi+26
2603.25339

Use fiducial model to analyse mis-specified mocks



- Ellipticity
- Anisotropy
- Baryons
- Mis-centring
- Observational uncertainties

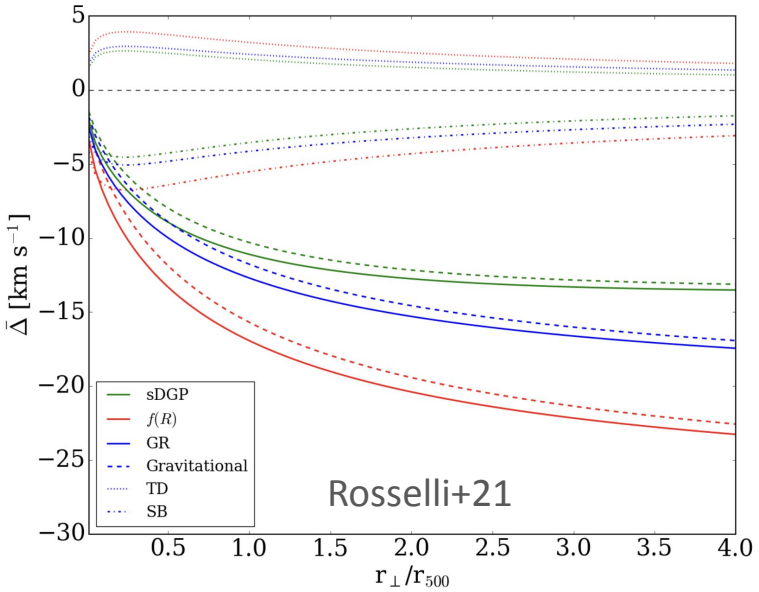
transverse Doppler > 0

$$\bar{\Delta} = \bar{\Delta}_{gz} + \bar{\Delta}_{TD} + \bar{\Delta}_{SB}$$

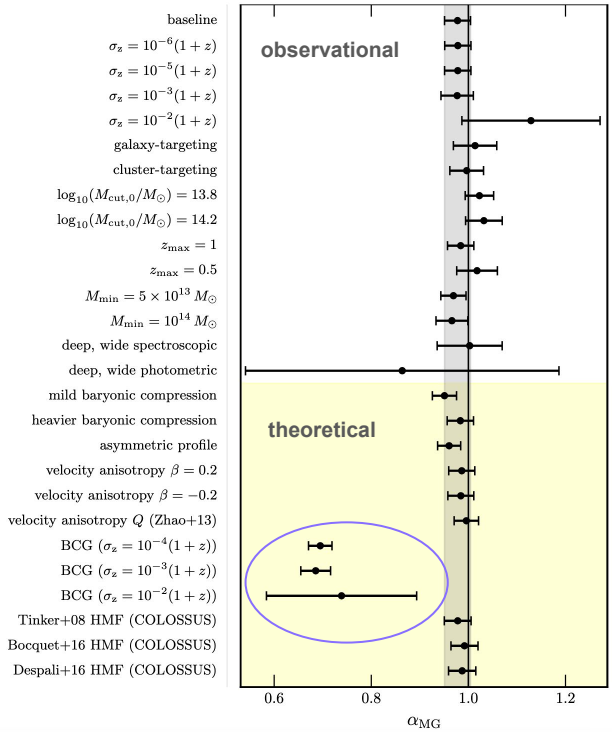
gravitational redshift < 0

surface brightness modulation < 0

Linear modified gravity parameter α_{MG}



Toward %-level constraints on gravity in the coming decade

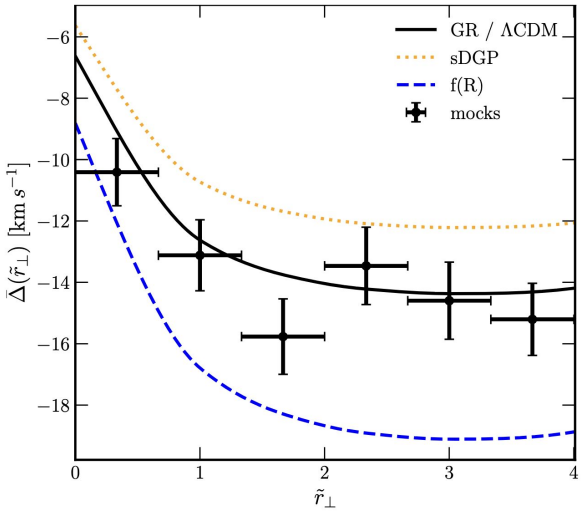


- Efficient setting: z < 1 spec-z galaxy survey
- 4% constraints on α_{MG} for m_{lim} = 22–23

Techniques to mitigate mis-centring



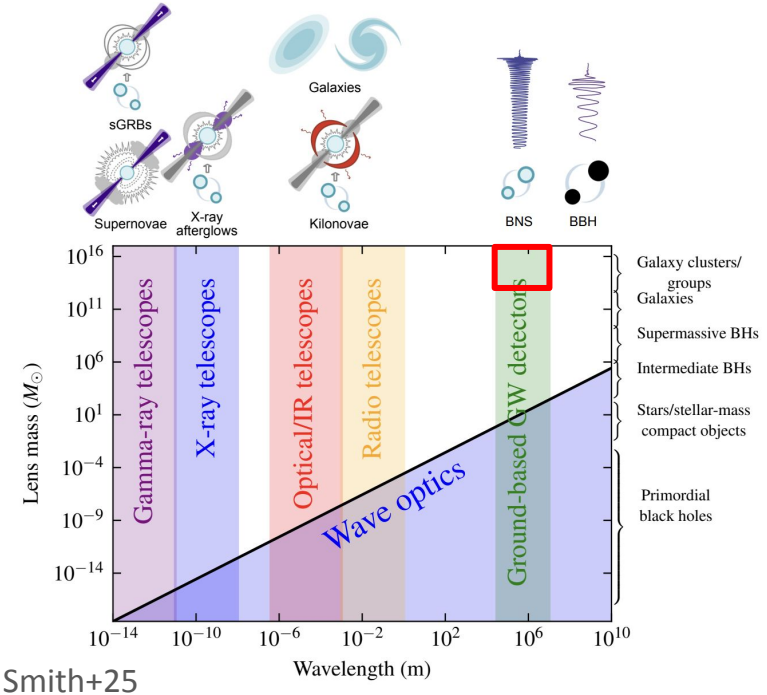
Tsaprazi+25
2512.13221



GW strong lensing: gravity at cluster scales

Tsaprazi+26
2609.24351

Signal: time delays between images



Smith+25

Cluster-lensed GWs:

- Exceptionally precise arrival times
- Can probe screening transitions
- Large cross sections
- Cluster forecast currently missing

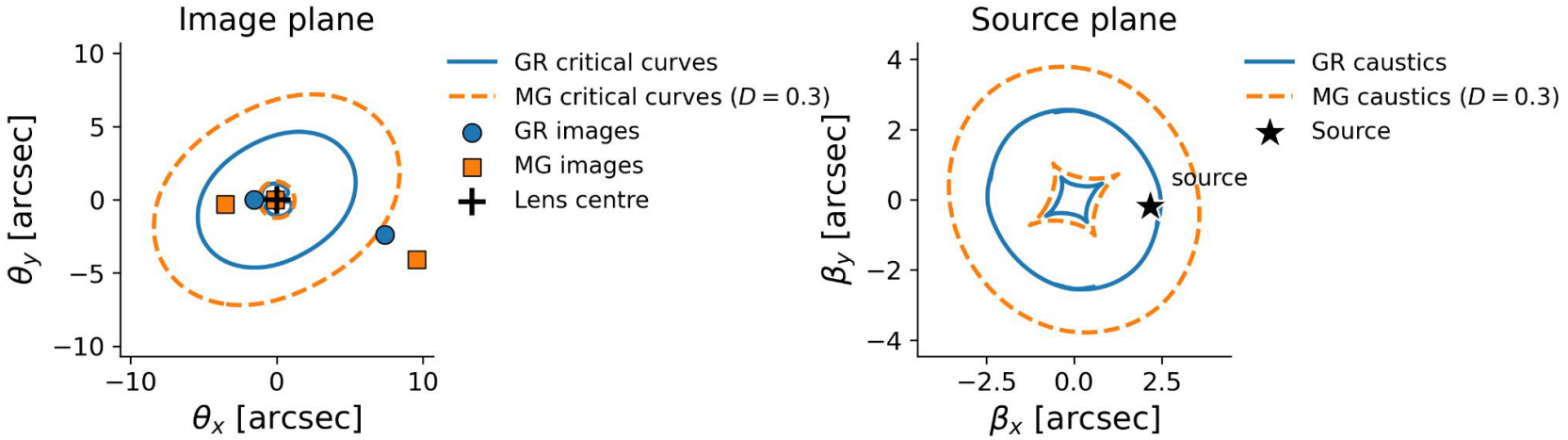
What are the modelling requirements for cluster-lensed GWs to provide robust tests of gravity?

Impact on strong lensing: geometry

$$\Sigma(r; D, A, R_{\rm scr}) = \begin{cases} 1\,, & 0 < r < R_{\rm scr}\,, \\ \dfrac{1 + \left(A + \dfrac{D}{2}\right)(r - R_{\rm scr})^2}{1 + \left(A - \dfrac{D}{2}\right)(r - R_{\rm scr})^2}\,, & r \geq R_{\rm scr}\,, \end{cases}$$

Reduced deflection angle

$$\boldsymbol{\alpha}(\boldsymbol{\xi}) = \frac{4G}{c^2}\frac{D_{\rm ls}}{D_{\rm s}} \int \mathrm{d}^2\xi'\, \Theta(\boldsymbol{\xi}')(\boldsymbol{\xi} - \boldsymbol{\xi}')I\big(|\boldsymbol{\xi} - \boldsymbol{\xi}'|; \Sigma\big)$$



Impact on strong lensing: time delays

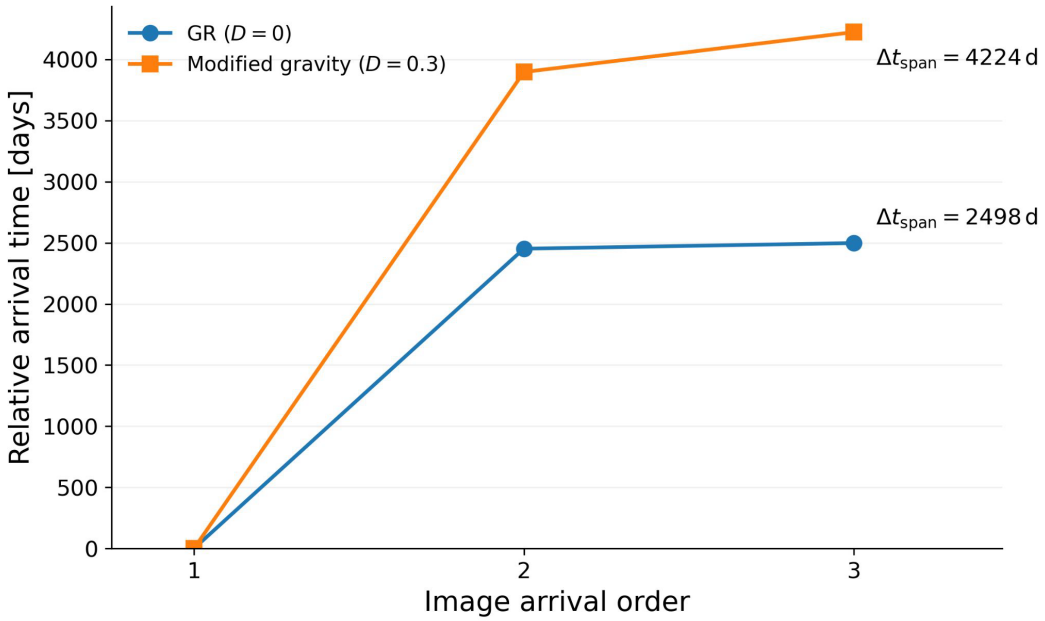
$$\Sigma(r; D, A, R_{\rm scr}) = \begin{cases} 1\,, & 0 < r < R_{\rm scr}\,, \\ \dfrac{1 + \left(A + \dfrac{D}{2}\right)(r - R_{\rm scr})^2}{1 + \left(A - \dfrac{D}{2}\right)(r - R_{\rm scr})^2}\,, & r \geq R_{\rm scr}\,, \end{cases}$$

Weyl potential

$$\Phi_+(r) = -\frac{G M_{\rm lens}}{r}\Sigma(r)$$

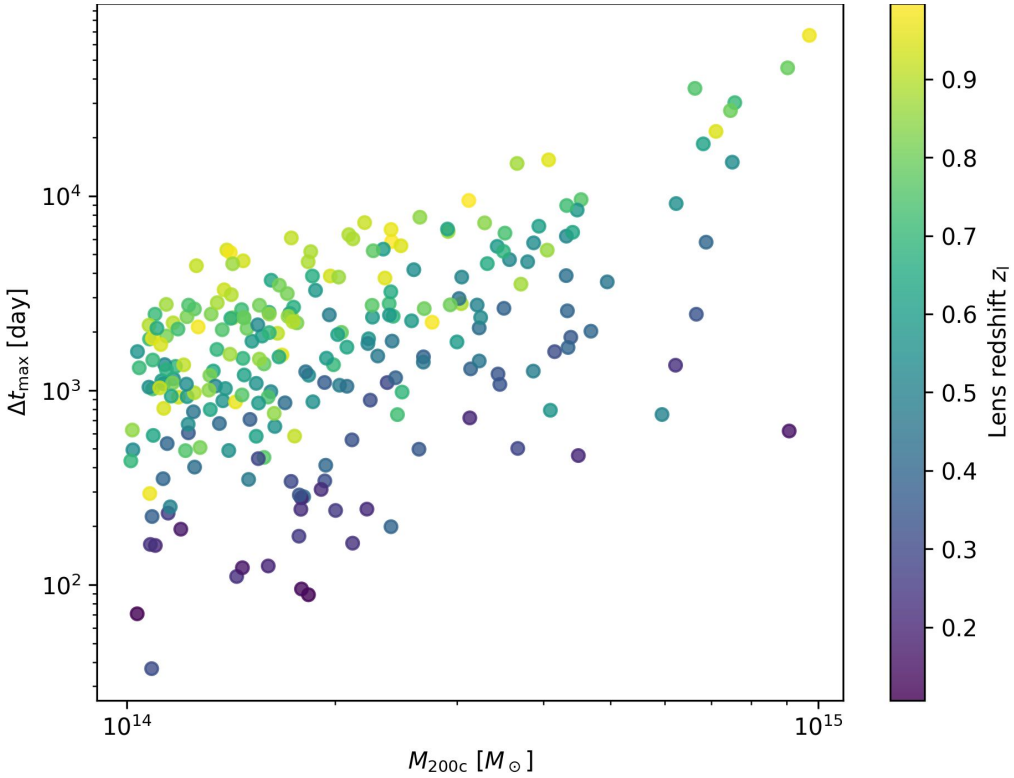
Time delays

$$\Delta t_{ij} = \frac{1 + z_{\rm l}}{c}\frac{D_{\rm l} D_{\rm s}}{D_{\rm ls}}\Delta\phi_{ij}$$

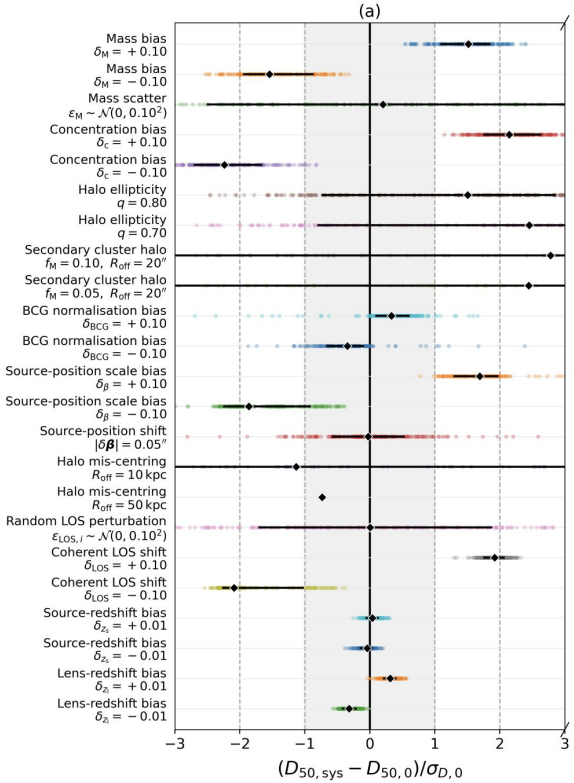


250 events sourced by binary neutron star mergers

- Source:
  - BNS
  - Position, z_s
- Lens:
  - Tinker HMF
  - NFW profile
  - BCG velocity dispersion
  - Asymmetry
  - Secondary haloes
  - Mis-centring
  - LOS structure
  - M_{200}, c_{200}, z_l



Lens modelling is crucial



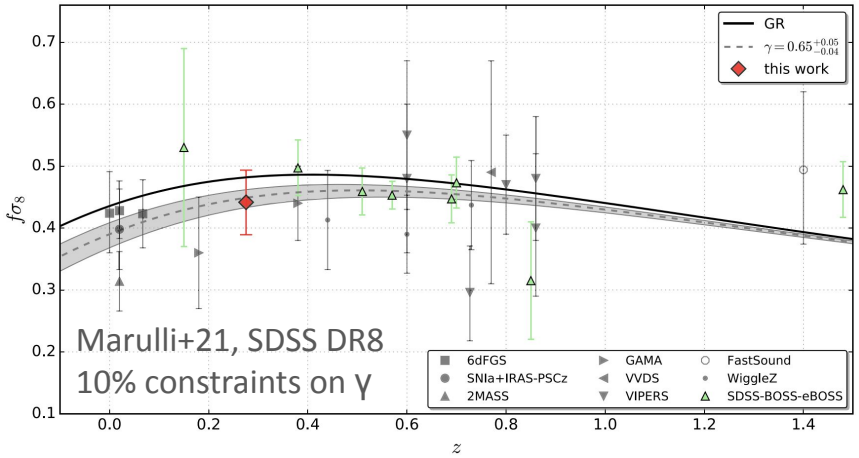
- **Lens-dependent** bias on D:
  - Ellipticity, secondary structure
  - LOS structure, mis-centring
  - Perturbed clusters should be avoided
- **Coherent** bias on D:
  - M_{200}, c_{200} biases
- Minimal effect:
  - BCG velocity dispersion, z_l, z_s

1. **Flexible forward modelling + external observations**
2. **Forecasts require hydrodynamical simulations**

Clustering: gravity at intercluster scales

- High-mass regime
- Breaks degeneracies with cluster counts
- Tight priors on tracer bias
- Probes the **growth index γ** through $f\sigma_8$

EC: **Tsaprazi**+26
2604.25762



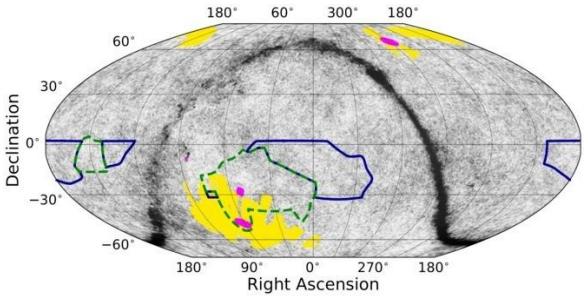
- **Anisotropic redshift-space power spectrum**
- Spectroscopic surveys: higher-order non-local bias + loop corrections
- Photometric surveys: Is the same complexity required?
- What about **Euclid DR1**?

A hierarchy of redshift-space clustering models

loop corrections
divergence
density

$$P^{\rm s}(k,\mu,z) = D[k\mu\sigma_{\rm v}(z)]\left[1+\frac{f(z)}{b(z)}\mu^2\right]^2 b^2(z)P_{\delta\delta}(k,z)$$

DR1-like 3LPT mocks



spectro-z

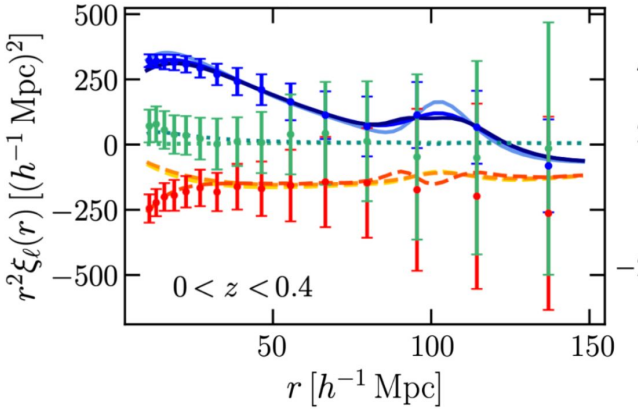
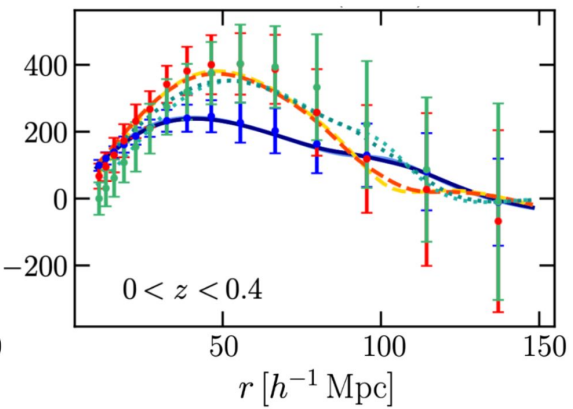
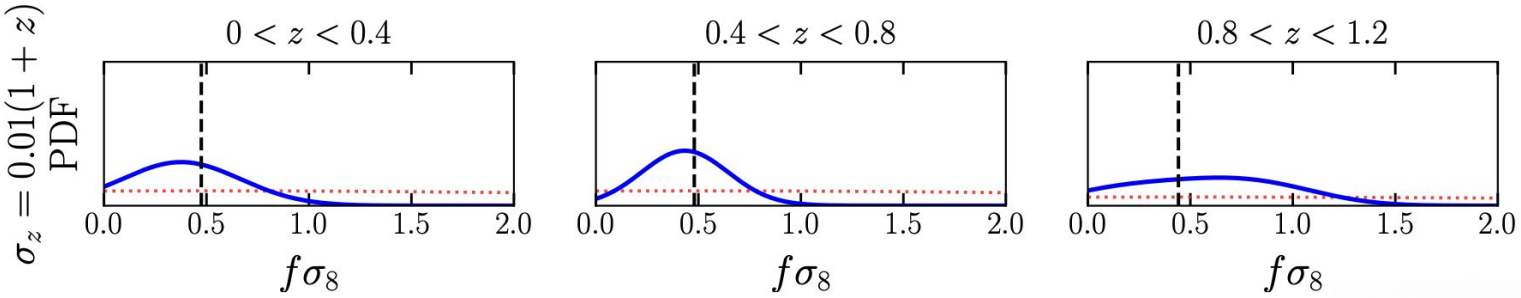


photo-z



- Different predictions on small scales, for spectro-z
- Photo-z: minor difference, down to what scales can we trust the simplest model?

Clustering: gravity at intercluster scales



- **Dispersion model – inference robust to clustering model > 10 h^{-1} Mpc**
- DR1 data: analysis is guiding the choice of $r_{min,\ell}$

cloe-org

1. **Non-linear bias at high-z**
2. **Forecasts with the Flagship N-body simulation**

From complementary probes to unified constraints

Gravitational redshifts

Strong GW lensing

Clustering

A multi-scale and multi-messenger framework for tests of gravity

Gravitational redshifts: forecast

GW lensing: mitigate model mis-specification

Clustering: DR1 data